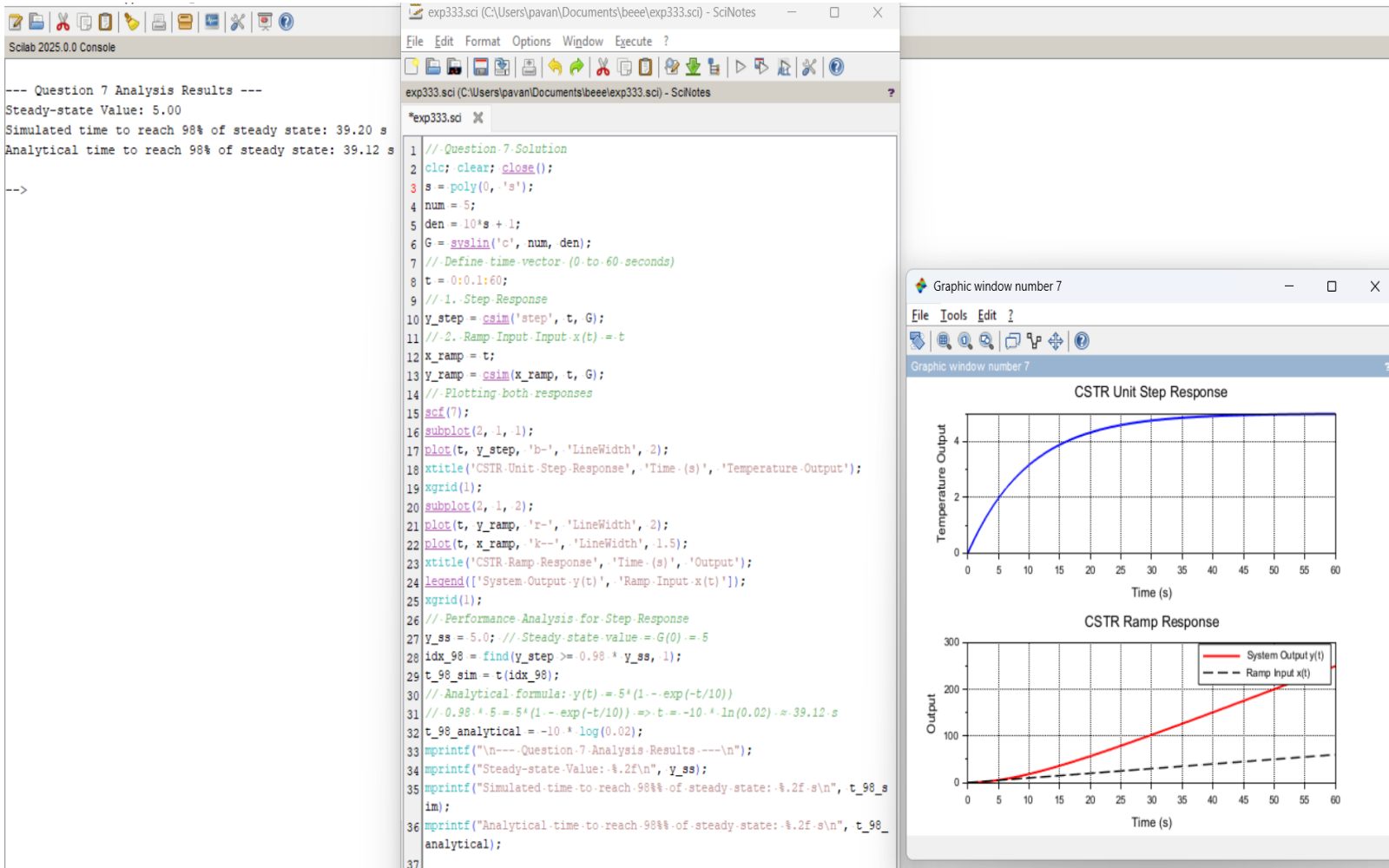


Question 7: CSTR Temperature Response (Step & Ramp Inputs)

Output:



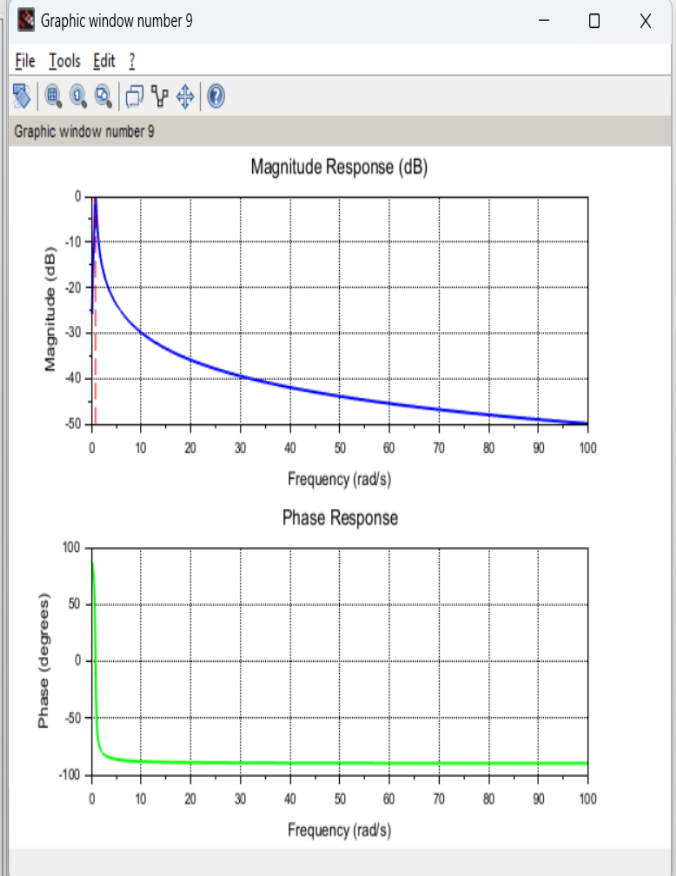
Question 9: Frequency Response & Bandwidth Extraction

--- Question 9 Analysis Results ---
 Center Frequency (Peak): 0.80 rad/s
 Lower -3 dB Frequency: 0.65 rad/s
 Upper -3 dB Frequency: 0.97 rad/s
 -3 dB Bandwidth: 0.32 rad/s

-->

```
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
*exp333.sci X

1 //Question-9-Solution
2 clc; clear; close();
3 s = poly(0, 's');
4 num = 2*s;
5 den = s^2 + 2*s + 25;
6 H = syslin('c', num, den);
7 w = logspace(-1, 2, 2000); //Frequency-vector-in-rad/s
8 [frq, rep] = repfreq(H, w);
9 mag_lin = abs(rep);
10 mag_dB = 20 * log10(mag_lin);
11 phase_deg = atan(imag(rep), real(rep)) * (180 / %pi);
12 //Locate-center-frequency-(Peak-of-|H(jw)|)
13 [peak_val, peak_idx] = max(mag_lin);
14 w_center = w(peak_idx);
15 peak_dB = mag_dB(peak_idx);
16 //Identify-3-dB-cutoff-frequencies
17 target_3dB = peak_dB - 3;
18 idx_lower = find(mag_dB(1:peak_idx) >= target_3dB, 1);
19 idx_upper = peak_idx + find(mag_dB(peak_idx:2000) <= target_3dB, 1) - 1;
20 w_low = w(idx_lower);
21 w_high = w(idx_upper);
22 bandwidth = w_high - w_low;
23 //Plotting
24 clf(9);
25 subplot(2, 1, 1);
26 plot(w, mag_dB, 'b-', 'LineWidth', 2);
27 plot([w_center, w_center], [min(mag_dB), peak_dB], 'r--');
28 xtitle('Magnitude-Response-(dB)', 'Frequency-(rad/s)', 'Magnitude-(dB)');
29 xgrid(1);
30 subplot(2, 1, 2);
31 plot(w, phase_deg, 'g-', 'LineWidth', 2);
32 xtitle('Phase-Response', 'Frequency-(rad/s)', 'Phase-(degrees)');
33 xgrid(1);
34 mprintf("\n--- Question-9-Analysis-Results---\n");
35 mprintf("Center-Frequency-(Peak): %.2f-rad/s\n", w_center);
36 mprintf("Lower--3-dB-Frequency: %.2f-rad/s\n", w_low);
37 mprintf("Upper--3-dB-Frequency: %.2f-rad/s\n", w_high);
38 mprintf("-3-dB-Bandwidth: %.2f-rad/s\n", bandwidth);
```



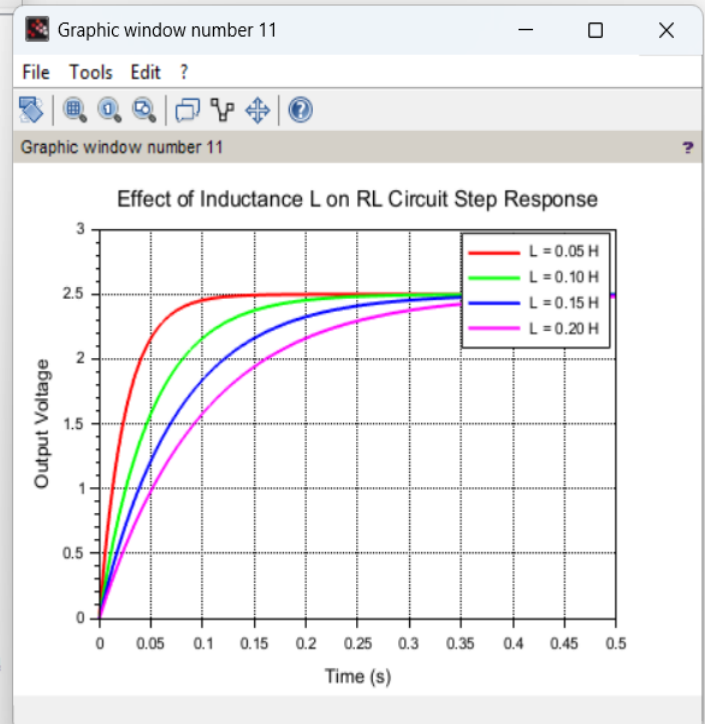
Question 11: Effect of Inductance on RL Circuit Step Response

--- Question 11 Analysis Results ---

L = 0.05 H | Time Constant (τ) = 0.025 s | Rise Time = 0.0550 s
L = 0.10 H | Time Constant (τ) = 0.050 s | Rise Time = 0.1100 s
L = 0.15 H | Time Constant (τ) = 0.075 s | Rise Time = 0.1650 s
L = 0.20 H | Time Constant (τ) = 0.100 s | Rise Time = 0.2200 s

```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
*exp333.sci X

1 //Question-11-Solution
2 clc; clear; close();
3 s = poly(0, 's');
4 R = 2;
5 K = 5;
6 L_list = [0.05, 0.10, 0.15, 0.20];
7 colors = {'r', 'g', 'b', 'm'};
8 t = 0:0.001:0.5;
9 scf(11);
10 mprintf("\n---Question-11-Analysis-Results---\n");
11 for i = 1:length(L_list)
12     L = L_list(i);
13     tau = L / R;
14     //G(s) = K ./ (L*s + R)
15     G = syslin('c', K, L*s + R);
16     y = csim('step', t, G);
17     //Measure rise time (10% to 90%)
18     y_ss = K / R; //2.5
19     i10 = find(y >= 0.1 * y_ss, 1);
20     i90 = find(y >= 0.9 * y_ss, 1);
21     t_rise = t(i90) - t(i10);
22
23     plot(t, y, colors(i), 'LineWidth', 2);
24     mprintf("L = %.2f H | Time Constant (tau) = %.3f s | Rise Time = %.4\n", L, tau, t_rise);
25 end
26 xtitle('Effect of Inductance L on RL Circuit Step Response', 'Time (s)', 'Output Voltage');
27 legend(['L = 0.05 H', 'L = 0.10 H', 'L = 0.15 H', 'L = 0.20 H']);
28 xgrid(1);
29
```



Question 14: Partial Fraction Decomposition & Impulse Response Comparison

--- Question 14 Analysis Results ---

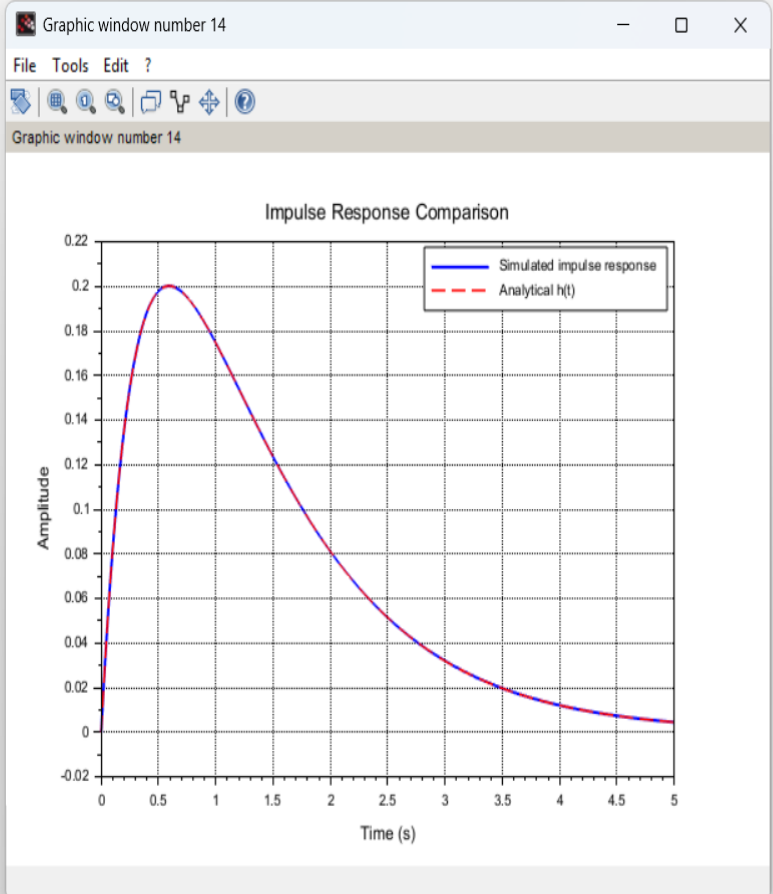
RMS Difference between Analytical and Simulated $h(t)$: 1.466628e-10

Dominant Time Constant: 1.00 s

```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
File Edit Format Options Window Execute ?

exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
*exp333.sci X

1 //Question-14-Solution
2 clc; clear; close();
3 s = poly(0, 's');
4 //Transfer-function H(s) = (s+3) / ((s+1)*(s+2)*(s+4))
5 num = s + 3;
6 den = (s + 1)*(s + 2)*(s + 4);
7 H = polyval('c', num, den);
8 //Partial-fraction-decomposition-using-pfss()
9 elts = pfss(H);
10 //Analytical-reconstruction-from-residues:
11 //H(s) = 2/3 * 1/(s+1) - 1/2 * 1/(s+2) - 1/6 * 1/(s+4)
12 t = 0:0.01:5;
13 h_analytical = (2/3)*exp(-t) - (1/2)*exp(-2*t) - (1/6)*exp(-4*t);
14 //Simulated-impulse-response-using-csim()
15 h_sim = csim('impulse', t, H);
16 //Calculate-RMS-difference
17 rms_diff = sqrt(mean((h_sim - h_analytical).^2));
18 //Dominant-time-constant-(smallest-magnitude-pole-is-at-s=-1)
19 tau_dominant = 1 / abs(-1);
20 //Plotting-comparison
21 scf(14);
22 plot(t, h_sim, 'b-', 'LineWidth', 2);
23 plot(t, h_analytical, 'r--', 'LineWidth', 1.5);
24 xtitle('Impulse-Response-Comparison', 'Time (s)', 'Amplitude');
25 legend(['Simulated-impulse-response', 'Analytical h(t)']);
26 xgrid(1);
27 mprintf("\n--- Question-14-Analysis-Results ---\n");
28 mprintf("RMS-Difference-between-Analytical-and-Simulated-h(t): %e\n", rms_diff);
29 mprintf("Dominant-Time-Constant: %.2f s\n", tau_dominant);
```



Question 18: Optical Channel Dispersion & Pulse Broadening

